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concentrator collects the data (voltage, current, etc) coming from all the micro-inverters and sends it to a local or remote monitoring and control access point.

The micro-inverter approach is highly preferable as it results in more energy production, smart communications and monitoring, and more flexibility and reliability.

Rising use of solar water heaters

In a day a family of four uses 5.12 units of electricity to heat a hundred litres of water, leading to an emission of 4.5 kg of carbon dioxide (CO₂). Installing a solar water heater (SWH) would result in a saving of 1500 units of electricity and a reduction of 1.5 tonnes of CO₂ emission annually, which is close to the burning of 650 litres of petrol.

"Awareness about the positive impact of SWH on financial and environmental aspects is enabling its foray into a wider range of application segments like individual houses, residential complexes (community SWH), hospitality, education institutes, HORECA and industries. In the new residential buildings, the government has mandated the use of SWH, which has helped improve awareness as well as usage in the residential condominium segment. Attractive payback period and future savings are making this technology popular," says Nitin Sangle, head-sales and service, solar, Racold Thermo.

There are lots of notable innovations in the SWH segment too. Here are some from Racold:

Omega flat-plate collectors. Equipped with a host of features, these collectors score over conventional solar flat-plate collectors. They have a specially-designed contact area that allows superior heat absorption, so you make the most of the sunlight! In a conventional collector, the fins

and risers are bound using ultrasonic or laser welding where the area of contact/heat transfer is 106 degrees. In the case of omega collectors, the area of contact between the fins and riser is 318 degrees, drastically increasing the contact area and hence maximising the heat transfer leading to higher efficiency.

High-energy absorption tube. Racold uses 1800mm long vacuum tubes, in keeping with international standards, ensuring optimal energy absorption and higher efficiency.

Sun Max reflector. Sun Max uses individualised V-shaped reflectors that ensure 85 per cent reflectivity of sun rays onto the tube, maximising product efficiency.

Synchrophasors help improve power system visibility and reliability

Power system disturbances may occur due to various system events, transmission congestions and fluctuations in renewable energy sources. A better understanding of these disturbances is essential to detect and prevent these events before they result in failures. Phasor measurement systems are used for this purpose.

Phasor measurement units (PMUs) are time-synchronised, high-speed, measurement units that monitor the current and voltage waveforms (sinusoids) in the grid, convert them into a phasor representation through

high-end computation, and securely transmit the same to a centralised server. When a phasor measurement is time-stamped against GPS universal time, it is called a synchrophasor. This allows measurements taken by PMUs in different locations or by different owners to be synchronised and time-aligned, then combined to provide a precise, comprehensive view of an entire region or interconnection.

"PMUs sample at speeds of 25 to 50 observations per second, compared to conventional monitoring technologies (such as SCADA) that measure once every two to four seconds and sometimes even once every five minutes," adds Mohanram.

A phasor measurement system typically includes high-end computation, parallel processing and modular, expandable input/output channels. It is rugged, field-worthy, upgradeable and capable of non-stop, real-time operation. It enables utilities to proactively plan energy delivery and prevent failures.

"A synchrophasor system is a wide deployment of PMUs and dedicated high-speed communication to collect and deliver synchronised high-speed grid condition data along with analytics and other advanced online dynamic security assessment and control applications. This will improve real-time situational awareness and decision support tools to enhance system reliability. Synchrophasor measurements can also be used to improve component and system models for both online and off-line network analysis to assess system security and adequacy to withstand expected contingencies. But to realise this great potential, each interconnection must deploy a highly reliable, secure and robust synchrophasor data measurement and collection system and develop a suite of validated, highly-available, robust and trustworthy analytical applications," explains Mohanram. The NI PMU is designed to meet the requirements of accuracy, reliability and interoperability; extreme environmental conditions, making it suitable for substation or